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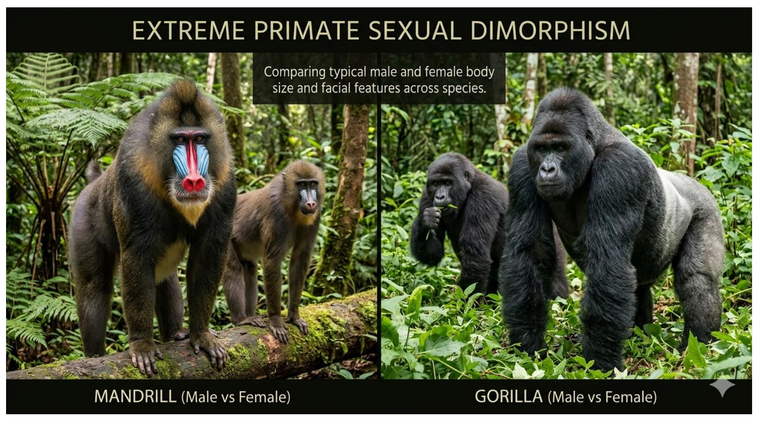
The Role of Evolvability and Integration in Shaping Sexual Dimorphism in the Cranial Morphology of Simiiform Monkeys

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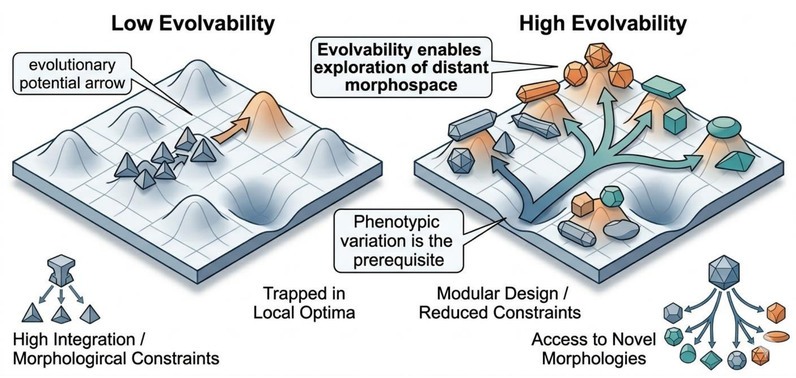
Sexual Dimorphism



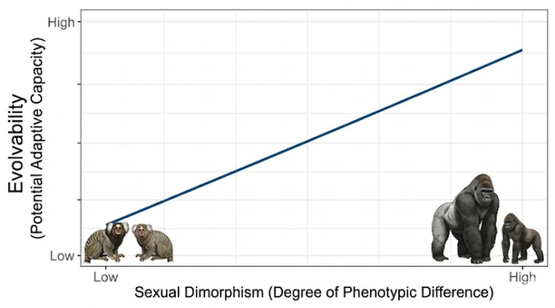
- Sexual dimorphism shows extensive interspecific variation
- The evolution of sexual dimorphism is likely multifactorial, but the role of evolvability remains unclear

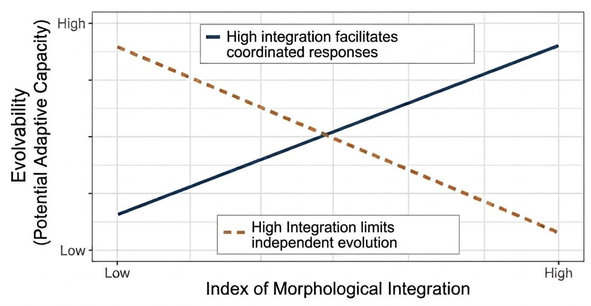
Evolvability and Integration

- Low evolvability may slow the pace of evolutionary change
- High evolvability may promote greater morphological diversification and faster evolutionary change



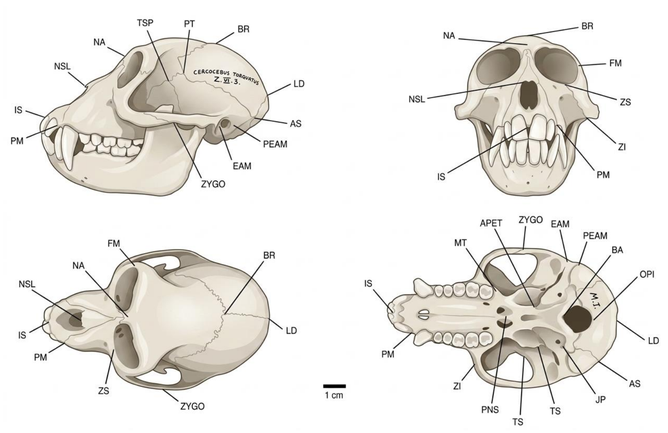
Hypothesis





Data

- 50 species from two simian clades: Platyrrhini and Catarrhini
- 38 linear cranial measurements collected from simian skulls
- ~ 5.416 simian skulls measured
- **P** matrices estimated for each species
- Sexual dimorphism was measured as the **norm** of the vector difference between males and females



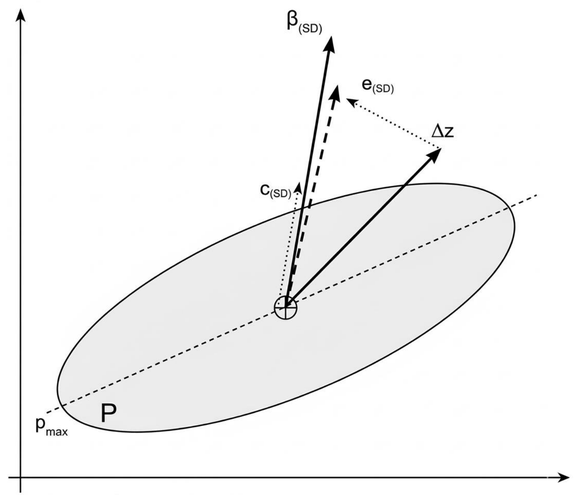
$$\text{Total Variation} = \text{Size (Pmax of Variation)} + \text{Shape (The Rest)}$$

$$\mathbf{P} \approx \mathbf{G}$$

(Phenotypic Correlation) ≈ (Genotypic Correlation)

$$\begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \bar{x}_3 \\ \vdots \\ \bar{x}_{38} \end{bmatrix}_{♂} - \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \bar{x}_3 \\ \vdots \\ \bar{x}_{38} \end{bmatrix}_{♀} = \begin{vmatrix} \text{SD} \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \bar{x}_3 \\ \vdots \\ \bar{x}_{38} \end{bmatrix} \end{vmatrix}$$

Analysis

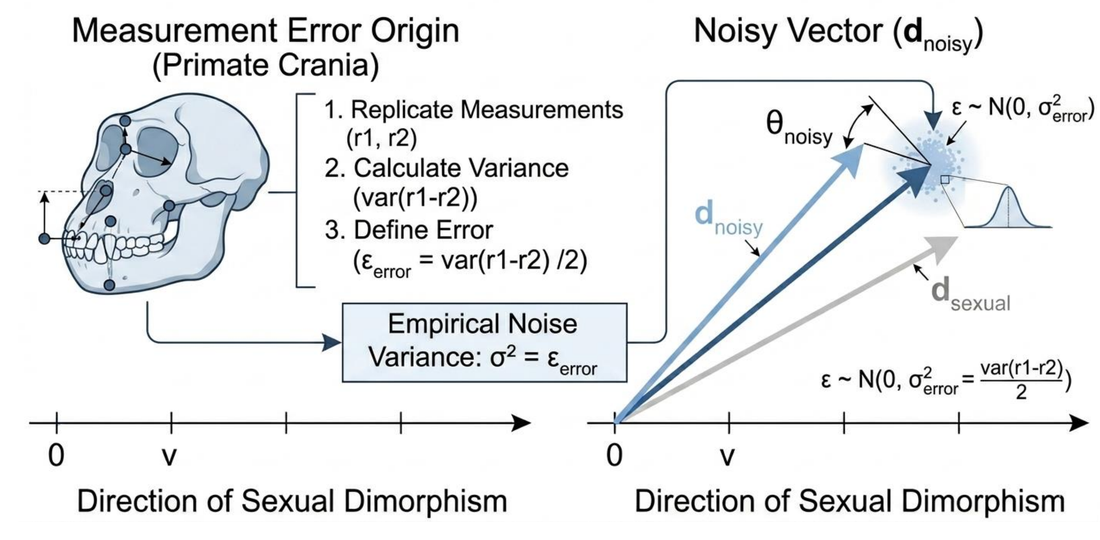




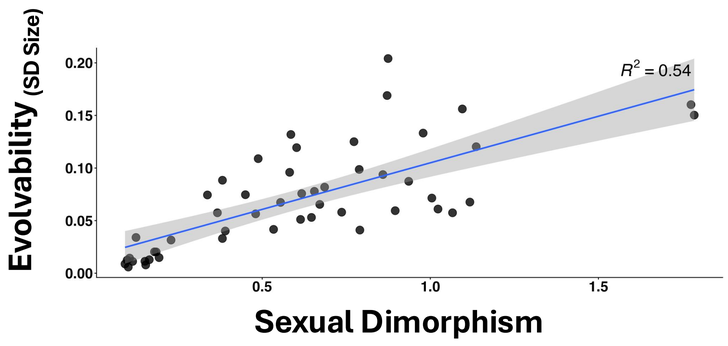
$$\text{evolvability}_{(SD)} = \frac{(SD + \varepsilon)' \mathbf{P}\, SD}{|SD + \varepsilon|^2}$$

$$\text{conditional evolvability}_{(SD)} = \frac{1}{SD^T \mathbf{P}^{-1} SD}$$

$$\text{integration} = 1 - \frac{\text{conditional evolvability}_{(SD)}}{\text{evolvability}_{(SD)}}$$

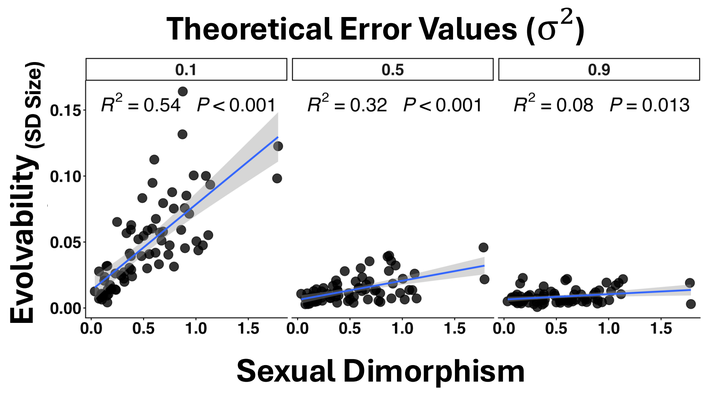


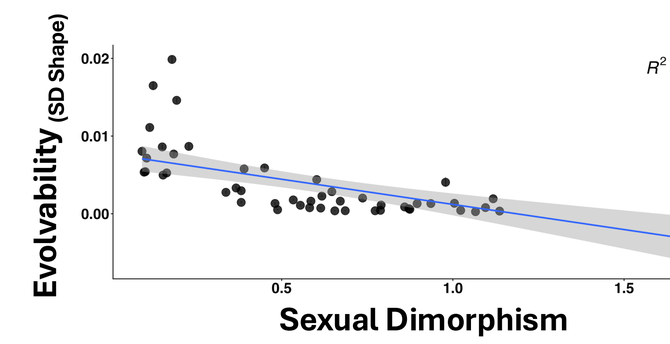
Results



- Greater sexual dimorphism is associated with higher evolvability in size

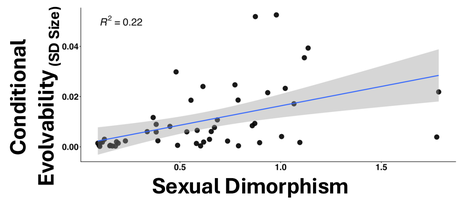
- High measurement error weakens and may eliminate the relationship between sexual dimorphism and evolvability

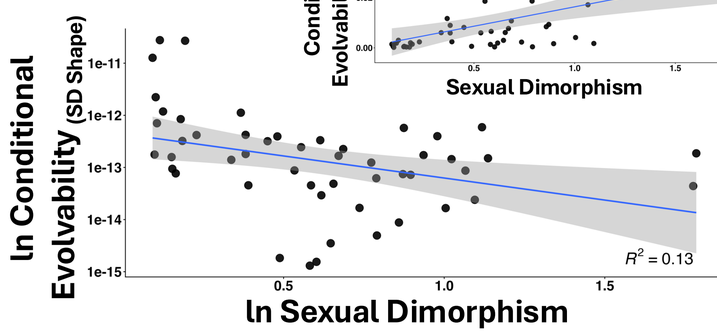


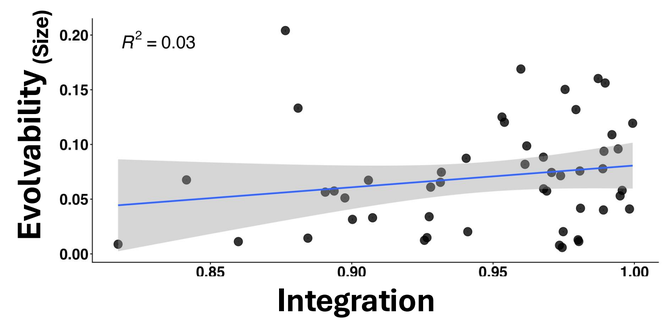


- The relation between sexual dimorphism and evolvability in shape is nonlinear and inverse

- Conditional evolvability of sexual dimorphism drops to practically zero after controlling for size







- No relationship was found between evolvability and integration
- Removing shape results in integration values close to the theoretical maximum (1)

Conclusions and Next Steps

- Sexual dimorphism tends to evolve along highly evolvable directions
- Integration alone does not influence evolvability among species
- Measurement error is important but does not substantially alter the observed patterns

- Test whether these patterns are phylogenetically widespread
- Better examine the consequences of removing size on evolvability estimates

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